DEVELOPMENT OF A NANO LUBRICATING OIL FROM AGRO WASTE FOR A DOMESTIC REFRIGERATION SYSTEM (TECHNICAL SUMMARY)

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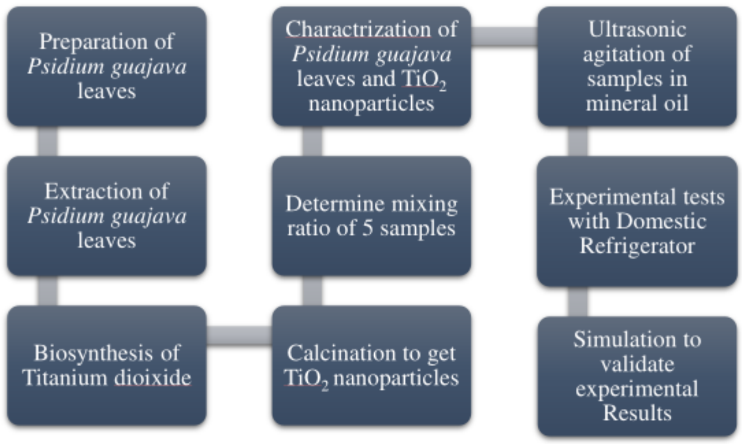
ABSTRACT

This study examines the creation, characterization, and performance assessment of a novel nano-lubricating oil derived from guava leaves for use in domestic refrigeration systems. This study's main goal was to examine the potential of guava leaf extract as an environmentally friendly additive in refrigeration lubricants and evaluate how it would affect system performance. The study started with the acquisition of guava leaf extract, which was then mixed in various concentrations—from 0.5 weight percent to 1.2 weight percent—with a polyester mineral oil. To ensure the uniform dispersion of the guava leaf extract within the base lubricant, homogenization was carried out using platform agitation. This process was designed to improve the compatibility and stability of the final nano-lubricating oil. Thorough characterization techniques were used to evaluate the make-up and characteristics of titanium dioxide nanoparticles and guava leaf extract. Insights into the surface morphology were gained through topological analysis, and the viability of the presence of titanium dioxide responsible for improving lubrication were found through chemical composition analysis. The characteristics of the titanium dioxide nanoparticles were also investigated to determine how they contributed to the overall performance of lubrication. Experiments were carried out in a table-top refrigerator to assess the performance of the nano-lubricating oil. In order to evaluate the system's efficacy and energy usage, the coefficient of performance (COP) was calculated. The ideal weight percentage of guava leaf extract in the lubricant was discovered through systematic testing and analysis to be 1 point 2 weight percent. The COP of the system was impressively increased by 24 percent at this concentration, demonstrating the enhanced lubrication capabilities offered by the guava leaf extract. The project's goals were met, it can be concluded, based on the results that were obtained. A sustainable and environmentally friendly replacement for traditional refrigeration lubricants has been demonstrated by the development of a nano-lubricating oil made from guava leaves. The performance of the system significantly improved as a result of the effective incorporation of the guava leaf extract at an optimized concentration, which also helped to increase energy efficiency and lessen the system's negative environmental impact. This study lays the groundwork for future investigation and advancement in the area of bio-based lubricants for refrigeration systems. Future research can look into additional plant extracts and nanoparticles to improve the sustainability and performance of lubricants, resulting in more effective and eco-friendly refrigeration technologies.

METHODOLOGY

From the chart below, the methodology followed through the process of the project.



CONCLUSION

This research made use of Psidium guajava in the production of TiO2 nanoparticles and utilized in a table-top refrigeration system. The results of the project are summarized as follows:

- TiO2 nanoparticles were successfully synthesized by reducing titanium tetra isopropoxide with guava leaf extract. The colour of the particles was black, representing the presence of impurities; carbon and indium.
- The composition of the developed TiO2 is on average 45.12wt% of Oxygen, 47.21wt% of titanium, 0.24wt% of carbon, and 1.29wt% of indium. Which shows a dominant presence of TiO2.
- The performance evaluation shows a deviation from the COP of the control by 24% by 1.2wt% concentration of TiO2 after 120 minutes of refrigeration operation.

The concentration of 1.2wt% is optimal from the experimental results.